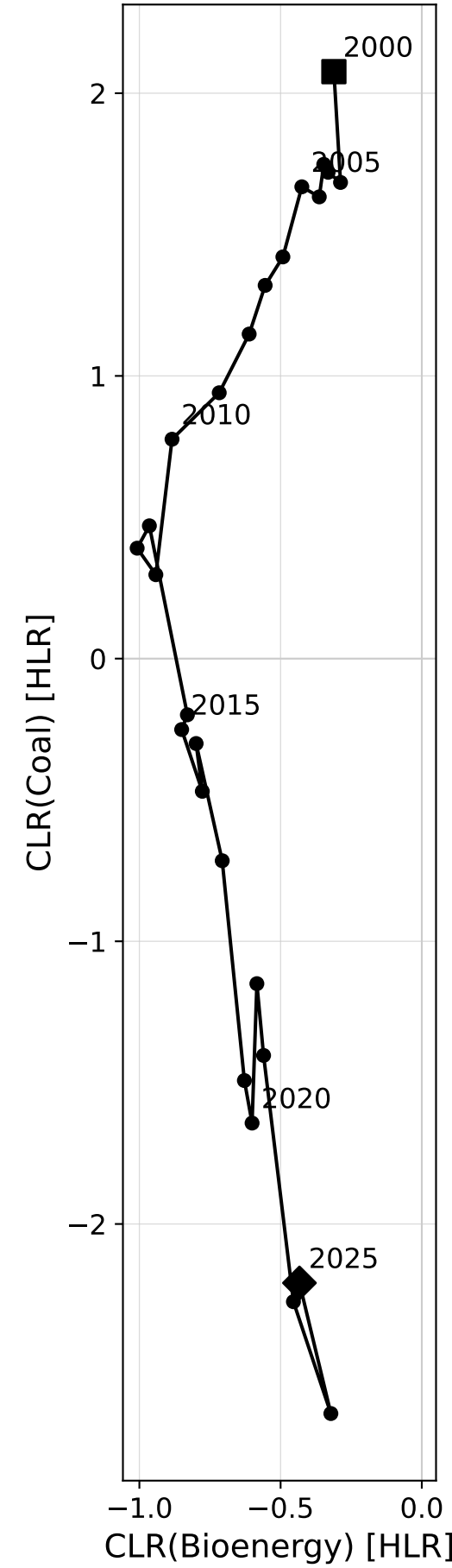
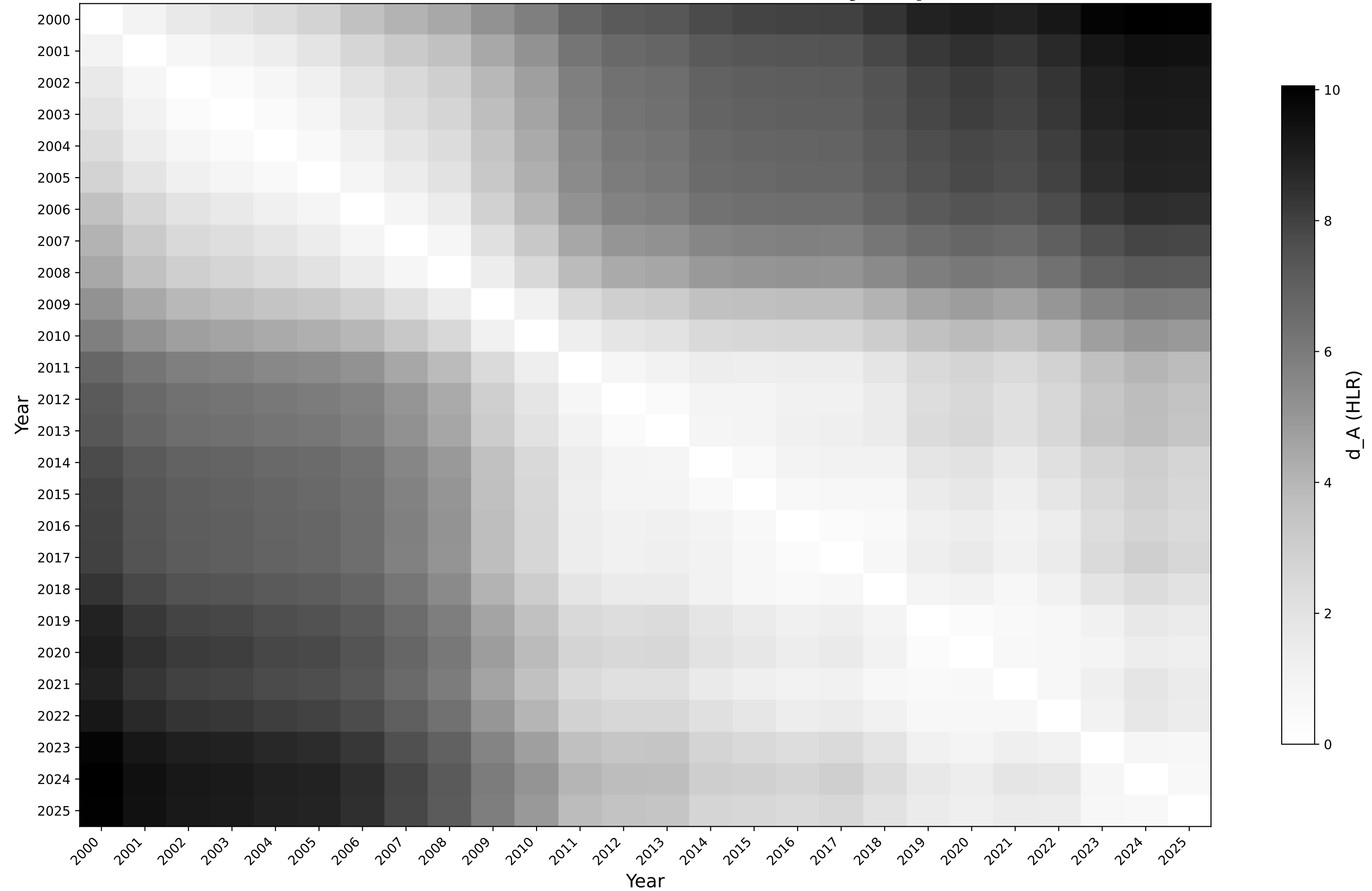


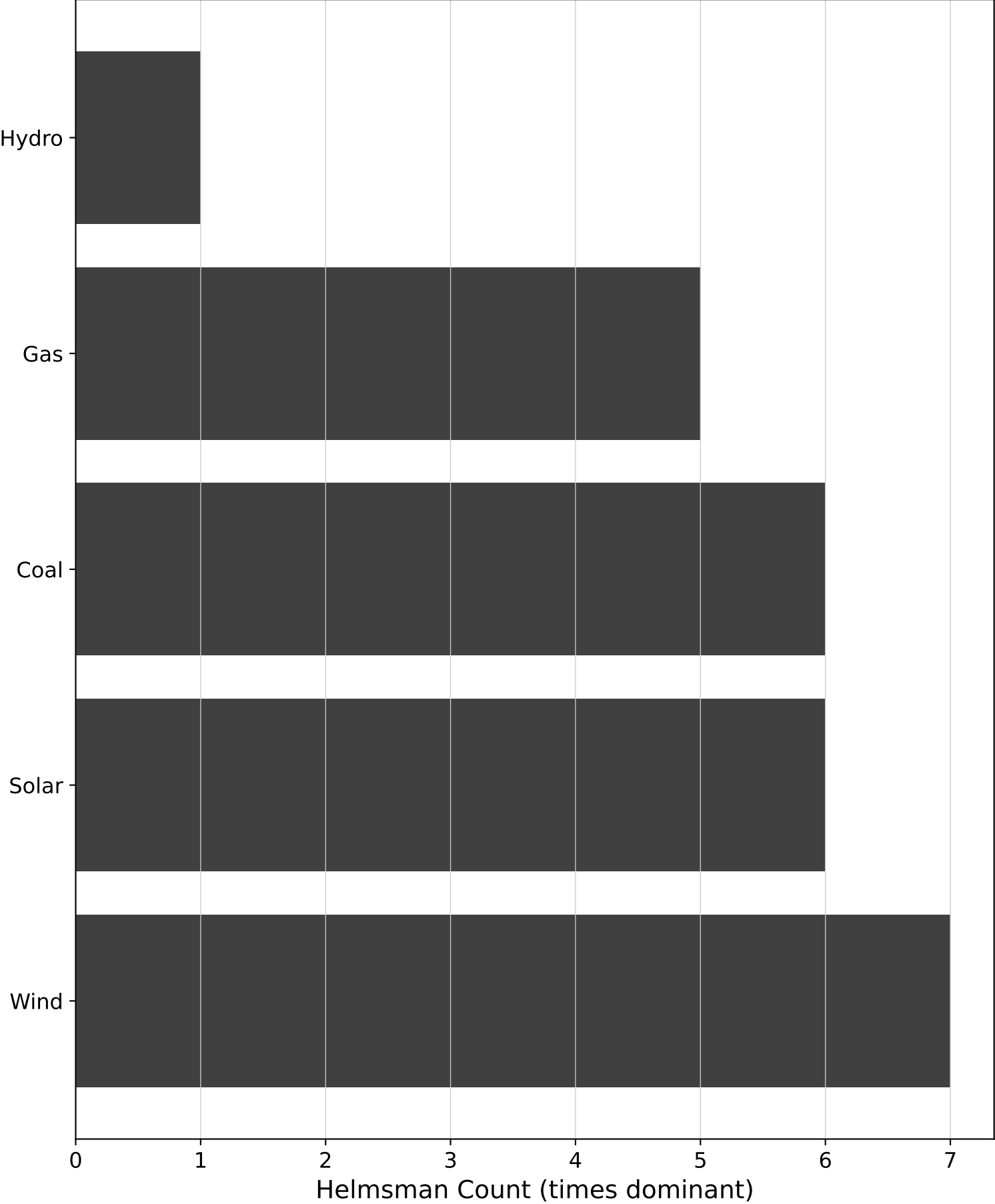
SYSTEM COURSE PLOT — XY Plan View (CLR Space)



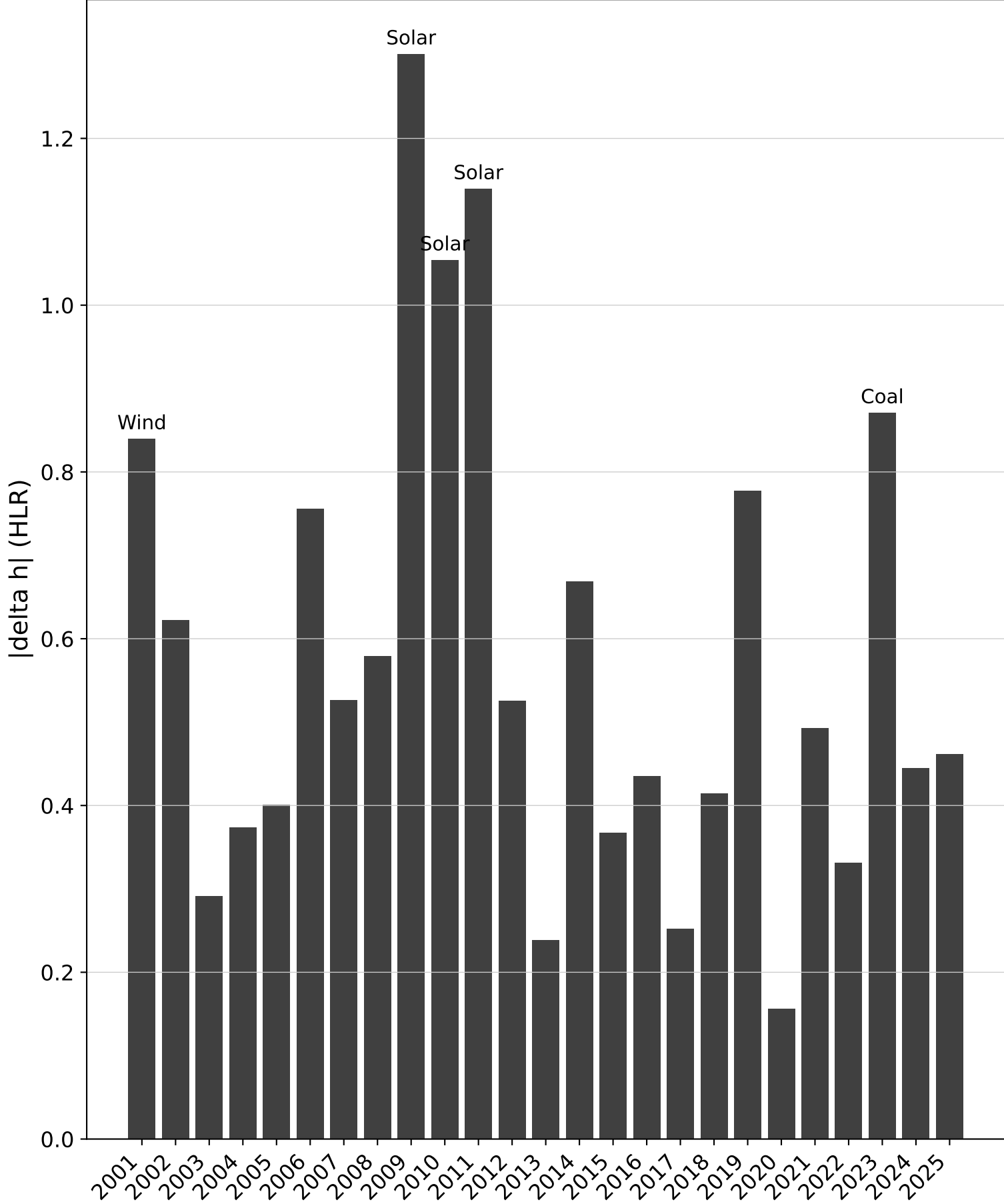
STAGE 2 — Aitchison Distance Matrix (all year pairs)



HELMSMAN FREQUENCY



HELMSMAN DISPLACEMENT PER YEAR



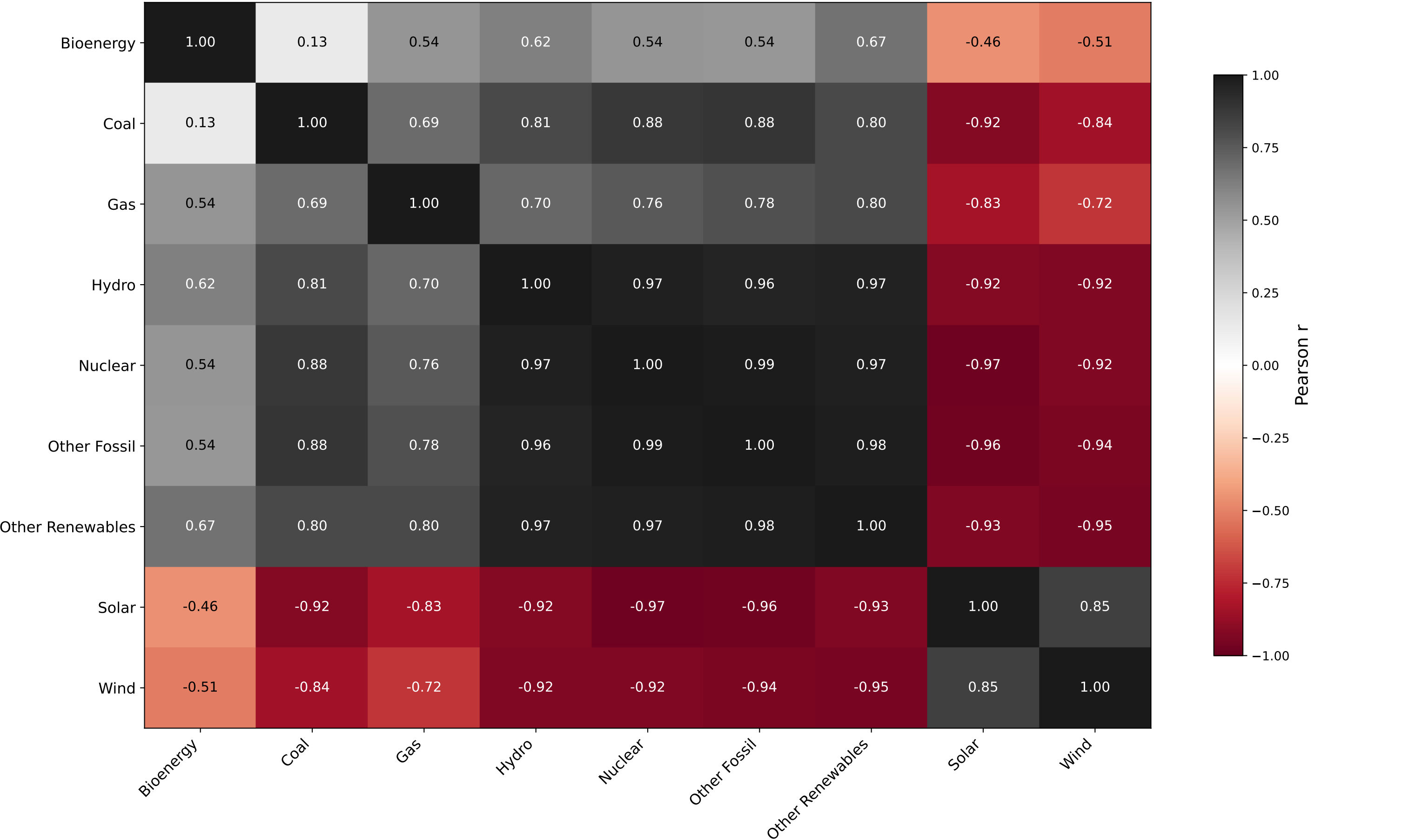
STAGE 2 — Pairwise Divergence Ranking (Top 15)

Rank	Year A	Year B	d_A (HLR)	Angle (deg)	Dominant Carrier
1	2000	2024	10.0592	76.5	Solar
2	2000	2025	10.0035	76.8	Solar
3	2000	2023	9.8631	75.4	Solar
4	2001	2024	9.4719	73.4	Solar
5	2001	2025	9.4392	73.9	Solar
6	2001	2023	9.2621	72.1	Solar
7	2000	2022	9.2488	69.7	Solar
8	2002	2024	9.2171	72.8	Solar
9	2002	2025	9.1780	73.2	Solar
10	2003	2024	9.1396	72.6	Solar
11	2003	2025	9.0974	72.9	Solar
12	2000	2020	9.0640	67.1	Solar
13	2002	2023	8.9766	71.1	Solar
14	2004	2024	8.9488	71.3	Solar
15	2004	2025	8.9135	71.7	Solar

STAGE 3 — Triadic Metric Triangle Area Ranking (Top 10)

Rank	Year A	Year B	Year C	Area (HLR^2)	Imbalance
1	2000	2007	2024	15.0050	2.47
2	2000	2007	2025	14.9541	2.46
3	2000	2006	2024	14.6841	2.83
4	2000	2006	2025	14.6763	2.81
5	2000	2007	2023	14.3006	2.42
6	2000	2008	2024	14.2167	2.27
7	2000	2008	2025	14.1315	2.26
8	2000	2006	2023	14.1174	2.77
9	2000	2008	2023	13.4178	2.22
10	2000	2007	2022	13.3358	2.27

STAGE 3 — Carrier CLR Trajectory Correlation



NAVIGATION SUMMARY — All Stage 1 Metrics by Year

Year	Hs	Ring	E_metric	omega(deg)	d_A	Helmsman	kappa_max
2000	0.6370	Hs-4	94.4	0.00	0.0000	---	53256
2001	0.6368	Hs-4	87.6	5.59	0.9978	Wind	54260
2002	0.6440	Hs-4	83.2	4.30	0.7323	Wind	55124
2003	0.6363	Hs-4	81.9	2.09	0.3386	Wind	55913
2004	0.6395	Hs-4	80.2	2.61	0.4205	Wind	56630
2005	0.6345	Hs-4	78.6	3.12	0.4937	Wind	56914
2006	0.6354	Hs-4	76.2	5.32	0.8274	Wind	56730
2007	0.6181	Hs-4	66.8	3.88	0.7982	Solar	28092
2008	0.6105	Hs-4	59.1	3.73	0.7104	Solar	14174
2009	0.6046	Hs-4	44.7	8.22	1.4367	Solar	3111
2010	0.5862	Hs-4	36.4	8.36	1.1295	Solar	1172
2011	0.5924	Hs-4	28.8	11.65	1.3373	Solar	1051
2012	0.5559	Hs-4	27.4	7.72	0.7241	Solar	1113
2013	0.5451	Hs-4	27.6	3.93	0.3603	Gas	1154
2014	0.5892	Hs-4	26.0	9.07	0.8332	Coal	1011
2015	0.5659	Hs-4	25.2	5.44	0.4868	Gas	990
2016	0.5185	Hs-4	24.9	6.10	0.5327	Gas	931
2017	0.5046	Hs-4	23.9	3.72	0.3381	Hydro	855
2018	0.5132	Hs-4	25.1	7.54	0.6621	Coal	945
2019	0.5019	Hs-4	26.8	9.80	0.8872	Coal	927
2020	0.4673	Hs-3	26.9	3.35	0.3035	Wind	864
2021	0.4789	Hs-3	25.6	6.20	0.5686	Coal	949
2022	0.4098	Hs-3	24.5	6.80	0.6039	Gas	781
2023	0.4442	Hs-3	29.5	10.88	1.0948	Coal	894
2024	0.4809	Hs-3	32.6	6.90	0.7272	Gas	986
2025	0.4838	Hs-3	30.0	5.50	0.5862	Coal	1000